Machine Learning Based Mental Health Prediction System for Early Detection of Depression and Stress Among Students

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Abstract

Mental health disorders such as depression, stress, and anxiety are increasing rapidly among students due to academic pressure, career uncertainty, social isolation, and unhealthy lifestyle habits. Early identification of mental health conditions is important because delayed treatment can negatively affect academic performance and emotional well-being. This paper presents a Machine Learning based Mental Health Prediction System capable of predicting mental health conditions using behavioral and psychological attributes. The proposed system analyzes user inputs such as sleep patterns, academic workload, mood swings, anxiety levels, and social interaction. Multiple Machine Learning algorithms including Logistic Regression, Decision Tree, Random Forest, and XGBoost are implemented and compared for prediction accuracy. Experimental analysis shows that ensemble learning techniques such as Random Forest and XGBoost provide better prediction performance and higher reliability compared to traditional approaches. The proposed system aims to support students and healthcare professionals by providing faster and more accessible mental health assessment through an intelligent web-based application.

Keywords

Mental Health Prediction, Machine Learning, Depression Detection, Stress Analysis, Random Forest, XGBoost, Artificial Intelligence, Student Healthcare.

Introduction

Mental health is one of the most important aspects of human well-being. In recent years, depression and stress among students have increased significantly due to examination pressure, academic competition, financial problems, and social expectations. Many students avoid seeking professional help because of fear, lack of awareness, or limited accessibility to mental healthcare services. Traditional diagnosis methods require expert consultation and consume considerable time and resources.

Machine Learning and Artificial Intelligence have introduced new possibilities in healthcare systems by enabling intelligent prediction and analysis of large datasets. Machine Learning algorithms can identify patterns in behavioral and psychological data and generate accurate predictions. The proposed system focuses on predicting depression and stress levels among students using multiple Machine Learning algorithms. The system provides faster analysis and helps in identifying individuals who may require mental health support at an early stage.

Literature Review

Several researchers have applied Machine Learning techniques in healthcare applications for predicting mental health disorders. Previous studies used algorithms such as Logistic Regression, Support Vector Machine, Decision Tree, and Random Forest for depression prediction. Some studies utilized social media data and wearable device information to identify emotional patterns and stress levels.

Although many existing systems achieved good prediction accuracy, several limitations still exist. Some systems depend heavily on social media behavior, while others suffer from poor scalability and low real-time prediction capability. Researchers found that ensemble learning algorithms such as Random Forest and XGBoost improve prediction accuracy by combining multiple decision models. Recent advancements in Artificial Intelligence have significantly improved predictive healthcare systems and enabled efficient early-stage diagnosis.

Problem Statement

Mental health disorders are difficult to identify during their early stages because individuals often hesitate to discuss emotional problems openly. Traditional healthcare systems require manual assessment and professional consultation, which may not always be accessible to students. There is a need for an intelligent automated system capable of predicting mental health conditions using behavioral and psychological data. The proposed system addresses this challenge by developing a Machine Learning based prediction model for early detection of depression and stress among students.

Proposed Methodology

The proposed system collects user information through questionnaires related to behavioral and psychological factors such as sleep hours, academic pressure, mood swings, social interaction, anxiety levels, and lifestyle habits. The collected data is preprocessed to remove missing values and convert categorical data into numerical form.

The processed dataset is divided into training and testing datasets. Multiple Machine Learning algorithms are trained and evaluated using performance metrics such as Accuracy, Precision, Recall, and F1-Score. The system compares the performance of different algorithms and selects the best-performing model for prediction.

The final prediction system is integrated into a web-based application developed using Flask. Users can provide input data through the user interface, and the trained model generates prediction results in real time.

Dataset Collection and Preprocessing

The dataset used in this project was collected from publicly available mental health datasets available on Kaggle and Open Source Mental Illness (OSMI) surveys. The dataset contains attributes such as age, gender, sleep patterns, work pressure, study hours, mood swings, anxiety level, social interaction, lifestyle habits, and mental fatigue.

Data preprocessing is an important step in Machine Learning because raw datasets often contain missing values and inconsistencies. The preprocessing steps include:

1. Removing duplicate records.
2. Handling missing values.
3. Encoding categorical variables.
4. Feature scaling and normalization.
5. Data cleaning and transformation.

These preprocessing techniques improve model performance and prediction accuracy.

Machine Learning Algorithms Used

1. Logistic Regression:
Logistic Regression is a supervised learning algorithm used for binary classification. It predicts the probability of a user experiencing depression or stress.

2. Decision Tree:
Decision Tree creates classification rules based on feature importance. It is easy to interpret and useful for identifying important mental health indicators.

3. Random Forest:
Random Forest is an ensemble learning algorithm that combines multiple decision trees. It improves prediction accuracy and reduces overfitting. Experimental results showed that Random Forest achieved high prediction performance.

4. XGBoost:
XGBoost is an advanced boosting algorithm that improves model accuracy by sequentially correcting prediction errors. It achieved the best performance among all implemented algorithms.

5. K-Nearest Neighbors (KNN):
KNN classifies data points based on similarity and proximity to neighboring data points. It is useful for identifying behavioral patterns in smaller datasets.

System Architecture

The system architecture consists of the following modules:
1. User Interface Module
2. Data Collection Module
3. Data Preprocessing Module
4. Machine Learning Model Module
5. Prediction Module
6. Result Visualization Module

Architecture Flow:
User Input → Data Preprocessing → Feature Extraction → Machine Learning Model → Prediction → Dashboard Visualization

Implementation

Frontend Development:
The frontend of the application was developed using HTML, CSS, and JavaScript to provide a simple and user-friendly interface.

Backend Development:
The backend was implemented using Python Flask framework. Flask handles user requests, data preprocessing, and communication with Machine Learning models.

Machine Learning Environment:
The Machine Learning models were trained using Python libraries such as Pandas, NumPy, Scikit-learn, and Matplotlib.

Model Evaluation:
The trained models were evaluated using performance metrics including Accuracy, Precision, Recall, and F1-Score.

Results and Analysis

The proposed system was evaluated using multiple Machine Learning algorithms. Experimental results demonstrated that ensemble learning algorithms achieved better performance compared to traditional classification methods.

Performance Comparison:

Logistic Regression - Accuracy: 81%
Decision Tree - Accuracy: 85%
Random Forest - Accuracy: 91%
XGBoost - Accuracy: 93%
KNN - Accuracy: 84%

Among all models, XGBoost achieved the highest prediction accuracy and F1-Score due to its advanced boosting mechanism. Random Forest also produced highly reliable results and reduced overfitting issues. The results indicate that Machine Learning techniques can effectively predict depression and stress levels among students.

Advantages of the Proposed System

1. Early prediction of depression and stress.
2. Faster and automated mental health assessment.
3. High prediction accuracy using ensemble learning algorithms.
4. User-friendly and accessible web-based interface.
5. Reduced dependency on manual diagnosis.
6. Scalable system suitable for educational institutions.
7. Helps healthcare professionals in decision-making.

Future Scope

The proposed system can be further improved using advanced Artificial Intelligence technologies. Future enhancements may include:
1. Deep Learning integration for improved accuracy.
2. AI chatbot support for mental health counseling.
3. Mobile application development.
4. Cloud deployment for scalability.
5. Wearable device integration for real-time monitoring.
6. Real-time emotion recognition using facial analysis.
7. Integration with healthcare systems and counseling platforms.

Conclusion

This paper presented a Machine Learning based Mental Health Prediction System for early detection of depression and stress among students. The system successfully predicts mental health conditions using behavioral and psychological attributes. Multiple Machine Learning algorithms were implemented and compared for prediction accuracy. Experimental analysis showed that XGBoost and Random Forest achieved the best performance. The proposed system provides an efficient, scalable, and intelligent solution for mental health prediction and early intervention. This research demonstrates the potential of Artificial Intelligence and Machine Learning in healthcare applications and student well-being support systems.

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